SG No. 98 D_4^{10} $I4_122$ [tetragonal]

* plus set: $+[0,\ 0,\ 0]$, $+\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}\right]$

Table 1: Wyckoff site: `4a`, site symmetry: `2.22`

No.	position	mapping
1	$[0,\ 0,\ 0]$	`[1,2,7,8]`
2	$\left[0,\ \frac{1}{2},\ \frac{1}{4}\right]$	`[3,4,5,6]`

Table 2: Wyckoff site: `4b`, site symmetry: `2.22`

No.	position	mapping
1	$\left[0,\ 0,\ \frac{1}{2}\right]$	`[1,2,7,8]`
2	$\left[0,\ \frac{1}{2},\ \frac{3}{4}\right]$	`[3,4,5,6]`

Table 3: Wyckoff site: `8c`, site symmetry: `2..`

No.	position	mapping
1	$[0,\ 0,\ z]$	`[1,2]`
2	$\left[0,\ \frac{1}{2},\ z+\frac{1}{4}\right]$	`[3,4]`
3	$\left[\frac{1}{2},\ 0,\ \frac{3}{4}-z\right]$	`[5,6]`
4	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}-z\right]$	`[7,8]`

Table 4: Wyckoff site: `8d`, site symmetry: `..2`

No.	position	mapping
1	$[x,\ x,\ 0]$	`[1,7]`
2	$\left[\frac{1}{2}-x,\ \frac{1}{2}-x,\ \frac{1}{2}\right]$	`[2,8]`
3	$\left[-x,\ x+\frac{1}{2},\ \frac{1}{4}\right]$	`[3,5]`
4	$\left[x+\frac{1}{2},\ -x,\ \frac{3}{4}\right]$	`[4,6]`

Table 5: Wyckoff site: `8e`, site symmetry: `..2`

No.	position	mapping
1	$[x,\ -x,\ 0]$	`[1,8]`
2	$\left[\frac{1}{2}-x,\ x+\frac{1}{2},\ \frac{1}{2}\right]$	`[2,7]`
3	$\left[x,\ x+\frac{1}{2},\ \frac{1}{4}\right]$	`[3,6]`
4	$\left[\frac{1}{2}-x,\ -x,\ \frac{3}{4}\right]$	`[4,5]`

Table 6: Wyckoff site: `8f`, site symmetry: `.2.`

No.	position	mapping
1	$\left[x,\ \frac{1}{4},\ \frac{1}{8}\right]$	`[1,6]`
2	$\left[\frac{1}{2}-x,\ \frac{1}{4},\ \frac{5}{8}\right]$	`[2,5]`

continued ...

Table 6

No.	position	mapping
3	$\left[\frac{3}{4},\ x+\frac{1}{2},\ \frac{3}{8}\right]$	`[3,7]`
4	$\left[\frac{3}{4},\ -x,\ \frac{7}{8}\right]$	`[4,8]`

Table 7: Wyckoff site: `16g`, site symmetry: `1`

No.	position	mapping
1	$[x,\ y,\ z]$	`[1]`
2	$\left[\frac{1}{2}-x,\ \frac{1}{2}-y,\ z+\frac{1}{2}\right]$	`[2]`
3	$\left[-y,\ x+\frac{1}{2},\ z+\frac{1}{4}\right]$	`[3]`
4	$\left[y+\frac{1}{2},\ -x,\ z+\frac{3}{4}\right]$	`[4]`
5	$\left[\frac{1}{2}-x,\ y,\ \frac{3}{4}-z\right]$	`[5]`
6	$\left[x,\ \frac{1}{2}-y,\ \frac{1}{4}-z\right]$	`[6]`
7	$\left[y+\frac{1}{2},\ x+\frac{1}{2},\ \frac{1}{2}-z\right]$	`[7]`
8	$[-y,\ -x,\ -z]$	`[8]`